MIDNIGHT BUILDATHON · WAVE 2

# A paper medical report anyone can check, without seeing the diagnosis.

A registered Nigerian doctor prints a QR code. A stranger scans it with no wallet and sees **Genuine**, **Revoked** or **Not on the registry**.

truereport.vercel.app

THE PROBLEM

# A Nigerian sick note is a letterhead, a stamp and a signature.

- The Embassy of Canada's verification officer: reports “lack security features” and “forgery is endemic” (IRB Canada, 2025).
- The Nigerian Senate asked the Health Ministry for a way “to stop the proliferation of fake medical reports” (2019).
- Checking one today means phoning the hospital. A shared database would fix it by leaking health data.

HOW IT WORKS

# Four steps. Only fingerprints reach the ledger.

### 1 · Register

The registrar adds hash(doctor key, name + MDCN folio) to a Merkle tree on Midnight.

### 2 · Issue

The doctor proves in zero knowledge their key is in the tree. Only a salted commitment and a revoke tag are written.

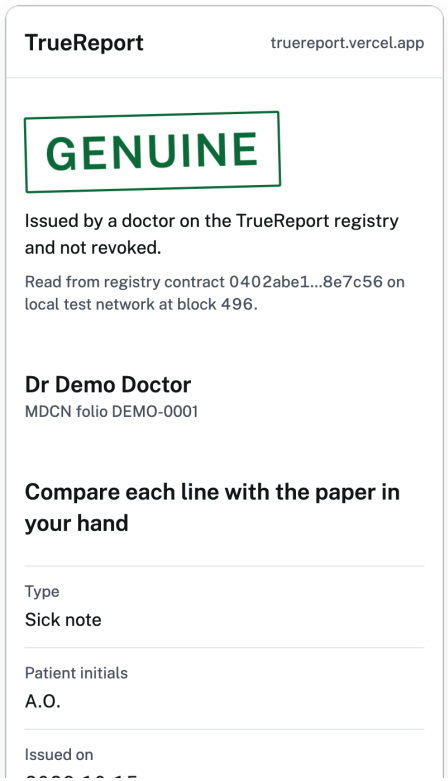
### 3 · Print

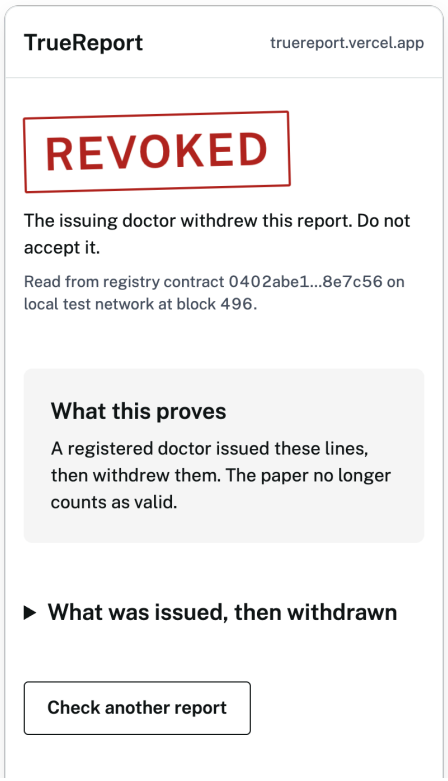
The QR carries the printed lines and the salt. Never the diagnosis text.

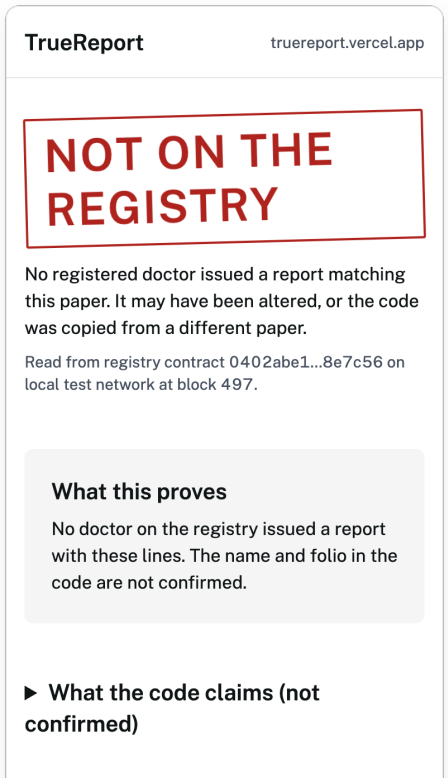
### 4 · Scan

The browser rebuilds the commitment and reads the ledger through the indexer. No wallet, no server of ours.

## WHAT A SCAN SHOWS




4 · captured from the real site on a local Midnight devnet

WHY MIDNIGHT

# A public registry that cannot say which doctor wrote which report.

- **issueReport**: ZK Merkle membership over a depth-10 tree; report hash, doctor id and salt stay private.
- **revokeReport**: tag = hash(key, commitment), so only the issuer can revoke and reports stay unlinkable.
- **Dual ledger**: keys, text and salts are private state; the tree, commitments and revocations are public.
- Without it: a public chain names the doctor on every report, or the verifier trusts our database.

WHAT IS REAL TODAY

| | |
|---|---|
| Contract: register, issue, revoke, refuse impostors | 21 of 21 simulator tests pass |
| Full flow on a Midnight network | Passed on a local devnet: add doctor, issue ×2, revoke, impostor refused, no-wallet read |
| Midnight Preview | Live: registry bc6440d9…69182b4, doctor + 2 reports, 1 revoked, impostor refused |
| Issue and revoke through Lace on the web | Built, not tested yet; the CLI is the tested path |
| That the medical opinion is correct | Not claimed: origin, integrity and revocation only |

WHO AND WHAT NEXT

# Built by a Nigerian medical intern who signs these notes.

- Issuers: licensed Nigerian doctors (MDCN: 58,000 renewed of 130,000 registered, 2024).
- Checkers: HR desks, embassies, insurers, courts. They need a phone camera, nothing else.
- 2026-10-22: show it to 5 doctors at my hospital. 2026-10-29: ask 3 HR desks if they would scan it.
- Wave 3: Lace issue/revoke tested, printable PDF template, a registrar flow for one NMA state branch.